

Advanced Principles of Neural Science
NSC 4330-01
Fall 2025
TTR, BSB C230, 12:30-1:45 p.m.
Joaquin N. Lugo Jr., Ph.D.
Professor

Office hours: Monday 10 a.m. to 1 p.m. or by appointment
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Texts Required

Neil Carlson. Foundations of Behavioral Neuroscience. 10th Edition.

Course Description:

Advanced Principles of Neural Science is an advanced survey of knowledge in the area of neuroscience as applied to behavior. The goal of this course is to examine multiple levels of the nervous system in detail. We will first study the topics surrounding the cellular aspects of the nervous system. We will then examine some of the methods used in behavioral neuroscience. The second half of the course will examine the role of neuroscience in sleep, emotion, ingestive behavior, learning and memory, and neurological disorders. Each core section will involve applying the information learned to current primary articles in neuroscience. The focus of both class discussions and laboratory is designed to examine the multiple levels of the nervous system.

Advanced Principles of Neural Science is an advanced course aimed at those students wishing to pursue graduate work in neuroscience or experimental psychology. Students who intend to pursue a career in clinical psychology, medicine, dentistry, physical or occupational therapy will also find this course a significant benefit.

Prerequisites:

Introduction to Neuroscience (NSC 1306) is the only official prerequisite required to enroll in this course. However, courses in chemistry, biology, and other advanced neuroscience courses (i.e., Drugs and Behavior, Clinical Neuroscience, Sensation and Perception, etc.) are strongly recommended. Advanced Principles of Neural Science is a capstone course in that it requires students to assimilate knowledge from many areas of neuroscience. At times, it will be necessary for students to draw on information gained in previous courses that is given only a cursory explanation by the text or the instructor.

Evaluation of Students:

Course grades will be based on three components: quizzes; tests; and comprehensive final exam.

Quizzes	20%
Exams	60%
Final Exam	20%

A = >89.5%	B+ = 87.5 to 89.4%	B = 79.5 to 87.4	C+ = 77.5 to 79.4
C = 69.5 to 77.4	D = 59.5 to 69.4	F = < 59.4	

Course grades. Course grades will be computed and posted occasionally during the semester: at the minimum grades are posted at midterm and after the final recitation day.

Cell phones, tablets, computers and ALL other electronic communication devices must be turned off and stored out of reach during ALL class periods.

Laboratory:

The laboratory in behavioral neuroscience is a separate class. You must be registered for NSC/PSY 4130 in addition to this course to receive laboratory credit.

Academic Integrity

Plagiarism or any form of cheating involves a breach of student-teacher trust. This means that any work submitted under your name is expected to be your own, neither composed by anyone else as a whole or in part, nor handed over to another person for complete or partial revision. Be sure to document all ideas that are not your own. In addition, you must not provide course materials to other students, whether individually or generally (such as online) that would enable them to gain an unfair academic advantage. Instances of plagiarism or any other act of academic dishonesty will be reported to the Honor Council and may result in failure of the course. Not understanding plagiarism is not an excuse. I expect you, as a Baylor student, to be intimately familiar with the [Honor Code](#).

Artificial Intelligence (AI)

Unless I instruct you otherwise, all assignments should be your original work and should not be produced in part or in total with the assistance of artificial intelligence (for example, ChatGPT, Grammarly, or some other resource). Use of artificial intelligence without explicit permission from the instructor constitutes a violation of the Honor Code at Baylor University. I will at times use artificial intelligence tools for instruction or allow or expect you to use them to complete your assignments, but in these instances I will provide clear permission and guidance for their proper use.

Academic Success

Center for Writing Excellence

The Division of Graduate Writing (DGW) offers free writing consultations for any writing project at any stage. Consultants are all experienced doctoral students from a variety of disciplines. The DGW Writing Center offers three appointment types: in-person consultations, video conferences, and written feedback. For in-person and video appointments, consultants will read your work for up to an hour ahead of your appointment and prepare feedback; you will then meet with them during your appointment time to discuss their feedback. For written feedback appointments, consultants will spend up to an hour reviewing your document and summarizing their feedback; written feedback appointments do not have a meeting component. The DGW also offers regular workshops related to writing, research, and publishing. Visit the Division of Graduate Writing page at cwe.artsandsciences.baylor.edu/consultations/dgw to find out more.

Accommodations

Any student who needs academic accommodations related to a documented disability should inform me immediately at the beginning of the semester. You are required to obtain appropriate documentation and information regarding your accommodations from the Office of Access and Learning Accommodation (OALA). Stop by the first floor of Sid Richardson, East Wing in the Paul L. Foster Success Center or call (254) 710-3605 or email OALA@baylor.edu.

Success Center

I believe every student who has been admitted to Baylor can be successful, and I want to partner with you to help you thrive academically. Be sure to take advantage of the many resources available for academic success, including coming to see me during my office hours. Students who regularly utilize the great resources in the [Paul L. Foster Success Center](#) such as [tutoring](#), [Learning Lab](#), and [Academic Mentoring](#) are among my most successful students. If your academic performance in this class is substandard, I will submit an Academic Progress Report to the Success Center so that the team of coordinated care professionals can ensure that you get the help you need.

First Generation College Students

Baylor University defines a first-generation college student as a student whose parents did not complete a four-year college degree. The First in Line program is a support office for first-generation college students to utilize if they have any questions or concerns. Please check out [First in Line](#), visit us in Sid Richardson, or email firstinline@baylor.edu.

Military Connected Students

Veterans, active-duty military personnel, and dependents are encouraged to connect with the [VETS](#) program, a space dedicated to supporting our military-connected students. Please communicate, in advance, if possible, any special circumstances (e.g., upcoming deployment, drill requirements, disability accommodations).

Mission

The mission of Baylor University is to educate men and women for worldwide leadership and service by integrating academic excellence and Christian commitment within a caring community.

Office Hours

One of the best ways to take full advantage of learning in my course is by coming to my posted office hours. I look forward to guiding you in your academic pursuits. Take advantage of the hours listed or email me for an appointment.

Student Wellbeing

I recognize that students face many struggles during their Baylor career. [The Department of CARE Team Services](#) is a group of dedicated and caring case managers who work with students facing mental health issues, financial struggles, and anything else affecting a student's ability to be successful in the classroom. You can find them on the second floor of the Student Life Center, suite 207. Contact Information: (254) 710-2100 ; CareTeam@baylor.edu.

Health Care

Baylor University is strongly committed to addressing the physical well-being and mental health of students by providing access to on-campus healthcare resources. [Baylor Health Services](#) includes Primary Care, Psychiatry, Physical Therapy and Pharmacy and is staffed with fully certified and licensed physicians and nurse practitioners, as well as nurses and administrative staff. Appointments may be made by calling their main number or by logging into the health portal located on their website. Contact Information: (254) 710-1010; Health_Services@baylor.edu.

Mental Health

You also have access to mental health services through [The Counseling Center](#). Connect with them for an initial appointment, goal-directed individual therapy, group therapy, nutritional counseling, and case management services. You can schedule your 30-minute initial appointment online through the [health portal](#) or by calling (254) 710-2467. If you are experiencing suicidal thoughts, call or text #988 anytime day or night for support.

Telehealth

All Baylor students also have access to medical and counseling services via telehealth in addition to the on-campus services. Medical services include virtual urgent care, psychiatry, and nutrition counseling. Counseling services include 24/7/365 in-the-moment support and ongoing mental health counseling. Access to care is available in the evenings, during weekends, and when the University

is closed. All services are free and unlimited to Baylor students. Please visit <https://baylor.academiclivecare.com/> for more information.

Addiction

[The Beauchamp Addiction Recovery Center](#) (BARC) supports students in recovery from substance and behavioral addictions through an all-encompassing level of support approach that includes one-on-one mentorship, support groups, and social events open to all Baylor students. Located in the East Village Residential Community (bottom floor of Teal Residential College). Contact Information: (254) 710-7092; BARC@baylor.edu

Food Insecurity

At Baylor, we want all students to have access to food resources that will support their holistic well-being and success. If you or someone you know experiences food insecurity at any time, you can find information on campus and community food resources by visiting [The Store](#). You can also contact Store staff at (254) 710-4931. For additional basic needs assistance, please reach out to [CASE](#) or the [Care Team](#).

Spirituality

[The Office of Spiritual Life](#) offers programs, persons, and resources to nurture theological depth, spiritual wholeness, and missional living. Located on the corner of 5th and Speight Street in the Bobo Spiritual Life Center. Contact Information: (254) 710-3517; Spiritual_Life@baylor.edu.

Sexual and Interpersonal Misconduct Policy and Civil Rights Policy

Baylor University does not tolerate unlawful harassment or discrimination on the basis of sex, gender, race, color, disability, national origin, ancestry, age, citizenship, genetic information or the refusal to submit to a genetic test, past, current, or prospective service in the uniformed services, or any other characteristic protected under applicable federal, Texas, or local law (collectively referred to as Protected Characteristics). These policies also prohibit discrimination and harassment based on pregnancy or related conditions.

If you or someone you know would like help related to an experience involving:

1. Sexual assault, sex-based harassment or discrimination, dating violence, domestic violence, stalking, sexual exploitation, or retaliation for reporting one of these types of prohibited conduct, please visit www.baylor.edu/titleix, or contact us at (254) 710-8454, or TitleIX_Coordinator@baylor.edu.
2. Harassment or discrimination (excluding those issues listed in #1) based on Protected Characteristics, please visit www.baylor.edu/civilrights, or contact us at (254) 710-8454 or Civil_Rights@baylor.edu.

For pregnancy or related conditions, the University will provide reasonable modifications to policies, practices, or procedures as necessary to ensure equal access to educational programs and activities. Individuals who need to request reasonable modifications should contact the [Office of Access and Learning Accommodation](#). More information on pregnancy and parenting resources can be found [here](#).

The Equity, Civil Rights, and Title IX Office understands the sensitive nature of these situations and can provide information about available on- and off-campus resources, such as counseling and psychological services, medical treatment, academic support, university housing, and other forms of assistance that may be available. Staff members at the office can also explain your rights and procedural options. You will not be required to share your experience. **If you or someone you know feels unsafe or may be in imminent danger, please call the Baylor Police Department (254-710-2222) or Waco Police Department (9-1-1) immediately.**

Except for Confidential Resources, all University Employees are designated Responsible Employees and thereby mandatory reporters of potential sexual and interpersonal misconduct violations. Confidential Resources who do not have to report include those

working in the Counseling Center, Health Center, Athletics Mental Health, and Baylor Psychology Clinic, and the University Chaplain and LHSON Chaplain.

Crises and Emergencies:

- **Counseling Center Crisis Line:** (254) 710-2467 (Business Hours/Non-Business Hours/Weekends)
- **Baylor Police Department:** (254) 710-2222
- **MHMR Crisis Center:** (254) 867-6550
- **MHMR 24-Hour Emergency/Crisis Number:** (254) 752-3451
- When home during academic breaks, when the counseling center is closed, please call your local resources and/or national hotlines:
 - National Hope Network Hotline: 1-800-SUICIDE (1-800-784-2433)
 - National Suicide Prevention Lifeline: 1-800-273-TALK (1-800-273-8255)

Religious Observances

I am committed to providing a caring environment in which students from diverse religious backgrounds can succeed. Please review all of the course requirements to identify assignments, exams, or attendance expectations that conflict with religious observance and related cultural practices. I ask that you communicate any conflicts to me within the first week of the semester to allow time for us to discuss the conflict and determine any fair and reasonable adjustments to the schedule and/or assignments. Keep in mind that some changes may simply not be possible for practical reasons of scheduling, but I welcome the opportunity to discuss this with you.

Baylor Land Acknowledgment

We respectfully acknowledge that Baylor University in Waco and its original campus in Independence are on the land and territories originally occupied by Indigenous peoples including the Waco and Tawakoni of the Wichita and Affiliated Tribes, the Tonkawa, the ~~Namunuu~~ (Comanche), Karankawa, and Lipan Apache. These Indigenous peoples were dispossessed of and removed from their lands over centuries by European colonization and American expansionism. In recognition that these Native Nations are the original stewards of Baylor's campus locations, the University strives to build sustainable relationships with sovereign Native Nations and Indigenous communities through education offerings, partnerships, and community service.

Classroom Atmosphere

Everyone deserves to be treated with respect and dignity in this classroom space: students, teaching assistants, instructors, and guests. Baylor's Code of Ethics ([BU-PP 024](#)) states, "Members of the Baylor University community are expected to act in a way that builds a distinctive sense of caring, kindness, mutual respect, collegiality, and fairness. The idea that the Baylor community is a family has always been vital in our historical traditions, and we here reaffirm our commitment to that idea. Out of respect for this community, we do not slander or defame." Incivility, unkindness, and aggression have no place in our classroom community.

Tentative Schedule

Date	Class	Week	Chap	Title
26-Aug	1	1	1	Origins of Behavioral Neuroscience
28-Aug	2	1	2	Structure and function of cells in the nervous system
2-Sep	3	2	2,3	Structure of the Nervous System Quiz 1 Chapter 2
4-Sep	4	2	3,4	Structure of the Nervous System, Psychopharmacology
9-Sep	5	3	4	Psychopharmacology Quiz 2 Chapter 4
11-Sep	6	3		Exam 1
16-Sep	7	4	5	Methods and Strategies of Research
18-Sep	8	4	5, 8	[1, 2] Quiz 3 Chapter 5, Control of Movement
23-Sep	9	5	8	Control of Movement
25-Sep	10	5	9	Sleep and Biological Rhythms Quiz 4 Chapter 8. Movement
30-Sep	11	6	9	Sleep and Biological Rhythms and Sleep article [3] Quiz 5 Chapter 9
2-Oct	12	6		Exam 2
7-Oct	13	7	10	Reproductive Behavior
9-Oct	14	7	10,11	Emotion Quiz 6 Chapter 10
14-Oct	15	8	11	Emotion [4]
16-Oct	16	8	12	Ingestive Behavior [5] Quiz 7 Chapter 11
21-Oct	17	9	12	Ingestive Behavior
23-Oct	18	9	13	Learning and Memory Quiz 8 Chapter 12
28-Oct	19	10	13	Learning and memory and PKM zeta papers [6, 7] Quiz 9 Chapter 13
30-Oct	20	10		Exam 3
4-Nov	21	11	14	Human Communication [8]
6-Nov	22	11	14	Neurological Disorders Quiz 10 Chapter 14
11-Nov	23	12	15	Neurological Disorders [9]
13-Nov	24	12	15	Neurological Disorders Quiz 11 Chapter 15
18-Nov	25	13	16	Schizophrenia, Affective Disorders, and Anxiety Disorders
20-Nov	26	13	16	Schizophrenia, Affective Disorders, and Anxiety Disorders Quiz 12 Chapter 16
25-Nov		14		No Class Thanksgiving
2-Dec	27	15	17	Autism, ADHD, and Substance Abuse, Prenatal Stress
4-Dec	28	15	17	Autism, ADHD, and Substance Abuse, Prenatal Stress [10] Quiz 13 Chapter 17
9-Dec	29	16		Exam Review
12-Dec		Final		Final Exam: Friday, December 12, 7:00-9:00 p.m.

I will place the following papers in the document folder before at least 1 class period before they are to be read. You are welcome to find them online and download them earlier. You can find them through Google scholar or Pubmed <https://pubmed.ncbi.nlm.nih.gov/>

1. Boyden, E.S., et al., *Millisecond-timescale, genetically targeted optical control of neural activity*. Nat Neurosci, 2005. **8**(9): p. 1263-8.
2. Ratti, E., et al., *Comparison of Medical and Consumer Wireless EEG Systems for Use in Clinical Trials*. Front Hum Neurosci, 2017. **11**: p. 398.
3. Adamantidis, A., M.C. Carter, and L. de Lecea, *Optogenetic deconstruction of sleep-wake circuitry in the brain*. Front Mol Neurosci, 2010. **2**: p. 31.
4. Koenigs, M., et al., *Damage to the prefrontal cortex increases utilitarian moral judgements*. Nature, 2007. **446**(7138): p. 908-11.
5. Zhang, Z., et al., *Actin filaments mediate mechanical gating during osmosensory transduction in rat supraoptic nucleus neurons*. J Neurosci, 2007. **27**(15): p. 4008-13.
6. Volk, L.J., et al., *PKM-zeta is not required for hippocampal synaptic plasticity, learning and memory*. Nature, 2013. **493**(7432): p. 420-3.
7. Tsokas, P., et al., *Compensation for PKM zeta in long-term potentiation and spatial long-term memory in mutant mice*. Elife, 2016. **5**.
8. Pulvermuller, F., et al., *Motor cortex maps articulatory features of speech sounds*. Proc Natl Acad Sci U S A, 2006. **103**(20): p. 7865-70.
9. Kaplitt, M.G., et al., *Safety and tolerability of gene therapy with an adeno-associated virus (AAV) borne GAD gene for Parkinson's disease: an open label, phase I trial*. Lancet, 2007. **369**(9579): p. 2097-105.
10. Salm, A.K., et al., *Lateral amygdaloid nucleus expansion in adult rats is associated with exposure to prenatal stress*. Developmental Brain Research, 2004. **148**(2): p. 159-167.